

Behavioral Verification Module - (BVM)

OOF™ Origin Open Foundation™

Independent Methodological Authority

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Architecture Ecosystem: Structured Reality Standards™

Architecture Family: AI Governance Architecture (AIG®)

Operational Layer: AI Behavioral Governance Layer

Governed Space: Behavioral Verification

Category: AI Governance

Subcategory: AI Behavioral Governance

Type: Behavioral Auditability Standard Module

Parent Standard: Behavioral Auditability Standard (BAS)

Version: 1.0

Status: Canonical · Open Module

Origin Date: 26 June 2026

Compatibility: OOF Methodology OS · AI Governance Architecture (AIG®) · Governance Architecture (GOA™) · Operational Reality Architecture (ORA™) · Cognitive Governance Intelligence Architecture (CLIA®) · Memory Governance Intelligence Architecture (MGIA™) · Accountability Governance Architecture (AGA™)

AI-Readable: Yes

Authority: OOF

Protection: MIP™ — Methodological Intellectual Property

Canonical Language: English (UCL)

Canonical Definition System

Canonical Definition

Behavioral Verification Module (BVM) defines the structural conditions under which AI behavior can be independently verified against governance policies, authorized objectives, operational constraints, behavioral evidence, and expected behavioral outcomes throughout the complete behavioral lifecycle.

BVM governs behavioral verification.

The module establishes the foundational conditions required to ensure that reconstructed AI behavior can be objectively confirmed as accurate, legitimate, governance-compliant, and operationally valid.

Reconstruction explains what happened.

Verification confirms whether it actually complied with governance.

BVM governs that verification.

Module Operational Role

BVM defines the behavioral-verification layer of BAS.

Its role is to confirm the validity, legitimacy, and governance compliance of reconstructed AI behavior.

Module Operational Space

BVM governs:

- behavioral verification
- governance verification
- behavioral validation
- policy verification
- behavioral compliance verification
- operational verification
- independent behavioral review
- verification governance

The module applies wherever reconstructed AI behavior must be independently verified before governance conclusions are reached.

Module Function

The module applies wherever systems must preserve:

- independently verifiable behavior
- governance-valid verification
- traceable verification results
- policy-compliant behavioral assessment
- objective behavioral confirmation
- trustworthy governance decisions

Its function is to ensure that reconstructed AI behavior is verified using objective governance evidence rather than assumptions.

Minimum Implementation Framework

1. Define the Behavioral Verification Object

The organization must define which AI behaviors require independent verification.

This may include:

- autonomous decisions
 - recommendations
 - Human-AI interactions
 - robotic actions
 - safety-critical behavior
 - financial decisions
 - escalation activities
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- multi-agent behavior

2. Define Verification Conditions

The system must define verification criteria.

This includes:

- governance policies
- behavioral objectives
- operational constraints
- compliance requirements
- behavioral evidence
- validation rules

3. Define Verification Failure Detection Logic

The system must identify verification failures.

This may include:

- unverifiable behavior
- policy conflicts
- incomplete evidence
- inconsistent behavioral history
- unsupported conclusions
- governance-invalid verification

4. Define Operational Response or Governance Logic

Governance response may include:

- verification review
- evidence reassessment
- behavioral investigation
- governance intervention
- compliance review
- operational restriction
- operational invalidation where required

5. Preserve Traceability & Restrict Invalid Conditions

The system must preserve reconstructable traceability of:

- verification activities
- governance reviews
- supporting evidence
- validation decisions
- intervention history
- resulting behavioral states

An AI behavioral environment must not remain auditability-valid if materially significant behavioral verification cannot be reconstructed, reviewed, validated, preserved, or governed.

Use Case 1 — AI Insurance Claims System

Scenario

An AI system automatically recommends approval or rejection of insurance claims.

Application

BVM verifies whether the AI behavior complied with governance policies, underwriting rules, regulatory requirements, and organizational procedures.

Result

The insurer strengthens governance confidence, improves regulatory compliance, reduces verification errors, and increases operational trust.

Use Case 2 — Autonomous Healthcare AI

Scenario

An AI clinical system recommends treatment options that later become subject to medical review.

Application

BVM verifies whether the behavioral recommendation complied with approved clinical protocols, governance policies, and authorized medical guidance.

Result

The healthcare organization improves governance transparency, strengthens patient safety, supports regulatory audits, and increases trust in AI-assisted clinical decision-making.

Canonical Closing Statement

Behavioral Verification Module (BVM) defines the structural conditions under which AI behavior can be independently verified against governance policies, authorized objectives, operational constraints, behavioral evidence, and expected behavioral outcomes throughout the complete behavioral lifecycle.

Behavior that can be reconstructed must also be verified. Behavioral Verification therefore becomes the second operational condition of Behavioral Auditability Standard within AI Governance Architecture (AIG®).

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Canonical Source

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